# Selection or adaptation under neoadjuvant PD-1 blockade

Bing Han · independent computational project · public data only · October 2026

## THE QUESTION

When a tumour looks different after treatment, two very different things can produce that difference. **Selection**: the cell populations were already present and their proportions changed, because treatment removed some and spared others. **Adaptation**: the same cell states persist and have changed what they express. Bulk measurements cannot separate them, and most single-cell differential expression is reported in a way that cannot either.

## DATA

GSE236581 — 22 colorectal cancer patients sampled before and after neoadjuvant anti-PD-1; 975,275 cells, 1.31 billion non-zero matrix entries, 212,862 tumour cells at paired timepoints. Validation cohort held back: GSE205506, 19 patients, independent group. Inclusion criteria and the screening of 453 GEO series were fixed before any dataset was opened.

## FINDING 1 — MOSTLY WITHIN CELL STATES, AND NOT SPECIFIC TO THE TUMOUR

Tumour epithelial cells fall from 37,804 to 22,416 while stromal cells double and T, B and myeloid cells rise. Despite that visible shift, the median per-patient composition share is **0.38** (IQR 0.33–0.48; 5 of 22 patients above 0.5; signed-rank against 0.5 $p = 0.0011$): about three fifths of the absolute gene-level change happens inside cell states, led by T cells in 12 of 22 patients and by epithelium in 8. It survives the granularity check pre-registered to break it — two annotation levels differ by **0.0245** against a 0.20 threshold.

**The pre-registered specificity control then failed.** The same pipeline on adjacent normal tissue, at the same timepoints, returns **0.36**. Two post-hoc follow-ups did not rescue it: the tumour does not change more in absolute terms (median ratio 1.14, 13 of 20 patients, $p = 0.23$), and the two tissues' per-gene changes do not agree within a patient beyond what a mismatched control produces. A real systemic effect, a generic property of paired single-cell timepoints, and shared pre-to-post batch drift all remain open; separating them needs paired timepoints with no treatment, which neither cohort has. **The measurement stands; the attribution does not.**

## FINDING 2 — A CELL-LEVEL TEST ON DATA CONTAINING NOTHING

Pre-treatment cells only, patients split at random 200 times, so no effect exists by construction. 15,716 genes tested; a calibrated test should call about 786 at $p<0.05$ and none after correction.

| Cells per patient | Cell-level, Bonferroni | Patient-level, Bonferroni | Replication cohort, cell-level |
|---|---|---|---|
| 100 | 165 | 0 | 153 |
| 300 | 951 | 0 | 1,023 |
| 1,000 | **3,324** | **0** | **4,394** |

The count tracks how many cells were pooled, not any signal: more cells buy more confidence about differences between patients that predate the treatment. Reproduced on the same curve in an independent cohort.

## METHOD COMMITMENTS

- Pre-registered before the counts matrix was downloaded, including three conditions that would make the result uninterpretable.
- The patient is the unit of evidence; every inference is across patients.
- The decomposition is an exact identity, asserted on every call, and refuses log input.
- No integration, no re-clustering, no trajectory inference — each excluded with its reason recorded.
- 39 unit tests; the matrix never lands on disk.

## WHAT IT DOES NOT CLAIM

- **Not a tumour-specific or treatment-specific result.** Adjacent normal tissue splits the same way and changes as much, and this cohort cannot say why.
- **Not replicated.** The validation cohort was opened once and the comparison proved impossible there — no cell-type annotation, and the pre-registration forbids deriving one. Recorded rather than worked around; the composition finding is permanently single-cohort.
- **Nothing about response.** Responder labels are not in the deposited data and were not inferred.
- **The labels are the authors'.** Agreement across granularities is within one annotation.
- **The audit's patient-level column is conservative**, because the exact test ignores ties — which flatters the comparison, so it is stated.

I designed the study and the statistical approach and interpreted the results; implementation was AI-assisted. Pre-registration, code and the record of claims my own controls refuted are available on request.